

Discrete Math W1

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ETH Zurich

Hello

I am Diego

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DM HS2026 - Group A

WhatsApp group



Plan

1. Introduction
2. Tips for your Studies
3. How to Study?
4. Plan for this Session
5. Theory Recap
6. Exercises
7. (Outlook)

Introduction

About the Course

Lectures: Mondays / Wednesdays 14:15 – 16:00 (ETA F5)

Exercise Sheets: Thursdays

The exercise session will cover the material from the previous week.

No written bonus exercises and no hand-ins.

Oral Bonus

Each student has 2 oral interviews in the semester.

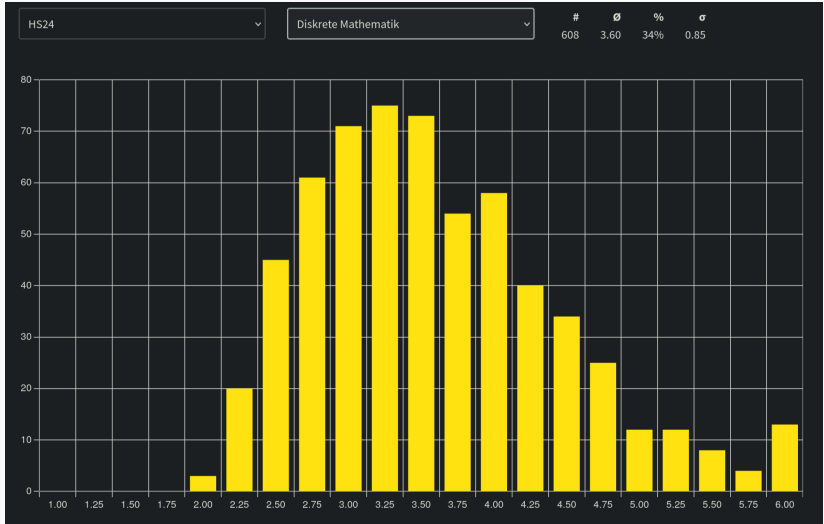
The main goal is to present your solution to the bonus problem of the current week.

The TA will interact with you to evaluate your understanding of the notions and give useful advice.

Maximum 6 points per interview.

If you're ill or can't attend, **you** have to organize a substitute
⇒ See Moodle

This is a hard course



BUT: It is possible!

DON'T: "This is so hard, I will never understand this."

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DON'T: "This is so hard, I will never understand this."

DO: "Actually, this isn't that complicated."

ASK QUESTIONS

The exam will be mainly proof-based $> 80\%$

So you really need to understand the material.

Don't fear to ask questions, you will almost never be the only one also having the same question.

How to Study?

Attend the lectures (in person if possible)

Attend the Ex. Sessions

Do the exercises

Ask questions

As you progress, you will also "learn to learn", so this can be very individual.

What I did

Read the script before the lecture. See the Roadmap.

Have a look at old exams

Write down on paper or tablet

Try to explain the topic to somebody

Discuss with other people (Not for the Bonus ex.)

If you have time, try all the exercises

I will give a (subjective) ordering of the exercises on the website

If you get stuck, wait, don't go to AI or the solution right away

AI can be helpful to get some hints

If you needed hints to solve an exercise, do it again a couple of days later by yourself

Math is NOT a SPECTATOR SPORT¹

Studying Mathematics is effective if it is a regular activity.

The only way to learn math is by doing it.

¹Inspired by Prof. Da Lio (Analysis III (Measure Theory))

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If you think you know something but don't write it down, you only think you know it.

— Leslie Lamport, Turing Award Winner

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AI can be a great learning partner

It will solve and explain every exercise

BUT...

it can be tempting to use it as a shortcut

"Failing is learning"

Plan for this Session

Structure (can change)

Theory Recap & Feedback (≈ 20 min)

Short Questions (≈ 25 min)

Break (≈ 15 min)

Exercises (≈ 35 min)

Questions & Outlook (≈ 10 min)

Extra Exercises

I will write an additional exercise sheet and publish it on my website.

Solutions will be provided $\approx$ 1 week later.

I don't expect you to solve all exercises.

Can be useful during exam prep.

If you don't like something, have suggestions, etc. Tell me, write me.

This should be **Your** exercise session.

This is my first time being a TA.

If I make a mistake, please tell me.

I really like (most) math and talking about it

If I ever go too fast, just tell me to slow down.

If you want to talk with me about other topics, don't hesitate, just make sure that it doesn't interrupt the session.

Questions?

Theory Recap

⇒ Basic Logic & Proofs

- Sets & Relations
- Number Theory
- Algebra
- Logic

1. Logic & Proofs

- Mathematical Statements
- Concept of a Proof
- Propositional Logic
- Predicate Logic
- Formulas vs. Statements
- Proof Patterns

A note on precision

This course is intended to teach you the necessary mathematical rigour.

Therefore, it focuses heavily on correct notation and precise statements.

If you're not precise enough, you will lose a lot of (unnecessary) points in an exam.

Definition 2.1

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- Every odd number greater than 5 can be expressed as the sum of three primes.
- There exists a polynomial-time algorithm for the shortest path problem in a graph.
- There are infinitely many primes of the form $2^p - 1$, where p is prime.

Definition 2.2

A **proof** of a statement S is a sequence of simple, easily verifiable, consecutive steps. The proof starts from a set of axioms (things postulated to be true) and known (previously proved) facts. Each step corresponds to the application of a derivation rule to a few already proven statements, resulting in a newly proved statement, until the final step results in S .

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Writing proofs is also a skill and can be learned.

Proofs (informally)

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Writing proofs is also a skill and can be learned.

Proofs can come in many forms.

If you want to know more, see *The Book of Proof* by Richard Hammack²

²Link on my website

Formal vs Informal Proof

Proof sketch

The non-obvious ideas used in the proof are described, but the proof is not spelled out in detail with explicit reference to all definitions that are used.

Complete Proof

The use of every definition is made explicit. Every proof step is justified by stating the rule or the definition that is applied.

Formal Proof

The proof is entirely phrased in a given proof calculus.

Exercises

Mathematical Statement or not?

$A =$ "Roger Federer is the best tennis player of all time."

$B =$ "Every composite number n has a prime divisor $p \leq \sqrt{n}$."

$C =$ "This statement is false"

$D =$ " $42 \in \mathbb{N}$ "

$E =$ "Some sets are finite"

$F =$ "The kernel of a homomorphism $\phi : R_1 \rightarrow R_2$ of rings is an ideal"

$G =$ "ETH Zurich is the best university"

Exercise 1.2

Find the mistake in the following proof.

Claim: 1 is the largest natural number.

Proof:

n is the largest natural number

$$\implies n^2 \leq n$$

$$\implies n(n-1) = n^2 - n \leq 0$$

$$\implies 0 \leq n \leq 1$$

$$\implies n = 1$$

$\implies$ vs. $\implies$ vs. $\rightarrow$

$\implies$ is used between two statements S, T

$S = "2^{64} - 2^{32} + 1 \text{ is prime}"$

$T = "Every \text{ even number greater than } 2, \text{ is the sum of } 2 \text{ primes}"$

$S \implies T$

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$\implies$ vs. $\Rightarrow$ vs. $\rightarrow$

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$S \implies T$

$\rightarrow$: is a logical operator on logical symbols

$\Rightarrow$: is to denote a step in a (formal) proof.

Logical Operators / Formulas

A	B	$\neg A$	$A \wedge B$	$A \vee B$	$A \rightarrow B$	$A \leftrightarrow B$
0	0	1	0	0	1	1
0	1	1	0	1	1	0
1	0	0	0	1	0	0
1	1	0	1	1	1	1

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0	0	1	0	0	1	1
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1	0	0	0	1	0	0
1	1	0	1	1	1	1

Definition 2.5

A correctly formed expression involving the propositional symbols $A, B, C, \dots$ and logical operators is called a **formula**.

Definition 2.6

Two formulas F and G (in propositional logic) are called **equivalent**, denoted as $F \equiv G$, if they correspond to the same function, i.e., if the truth values are equal for all truth assignments to the propositional symbols appearing in F and G .

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Definition 2.7

A formula G is a **logical consequence** of a formula F , denoted

$$F \models G$$

if for all truth assignments to the propositional symbols appearing in F or G , the truth value of G is 1 if the truth value of F is 1.

Basic Laws of Propositional Logic

Lemma 2.1.

- 1) $A \wedge A \equiv A$ and $A \vee A \equiv A$ (idempotence);
- 2) $A \wedge B \equiv B \wedge A$ and $A \vee B \equiv B \vee A$ (commutativity of $\wedge$ and $\vee$);
- 3) $(A \wedge B) \wedge C \equiv A \wedge (B \wedge C)$ and $(A \vee B) \vee C \equiv A \vee (B \vee C)$ (associativity);
- 4) $A \wedge (A \vee B) \equiv A$ and $A \vee (A \wedge B) \equiv A$ (absorption);
- 5) $A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$ (first distributive law);
- 6) $A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$ (second distributive law);
- 7) $\neg\neg A \equiv A$ (double negation);
- 8) $\neg(A \wedge B) \equiv \neg A \vee \neg B$ and $\neg(A \vee B) \equiv \neg A \wedge \neg B$ (de Morgan's rules).

Truth tables

Useful for proving statements about relatively short formulas.

Reduce longer formulas with Lemma 2.1

Use the convention for constructing the table (00, 01, 10, 11, etc.)

It does not suffice to only draw the truth table.

Make sure to explain where in it the result / counterexample can be seen.

Exercises

Exercise 1.5

Exercise 1.5 Two New Logical Operators

We define two binary logical operators $\heartsuit$ and $\diamond$ as follows:

A	B	$A \heartsuit B$
0	0	1
0	1	1
1	0	0
1	1	1

A	B	$A \diamond B$
0	0	0
0	1	1
1	0	1
1	1	0

1. (★) Are $\heartsuit$ and $\diamond$ commutative, i.e. does it hold

$$A \heartsuit B \equiv B \heartsuit A \quad \text{and} \quad A \diamond B \equiv B \diamond A?$$

Argue by comparing function tables.

Exercise 1.6

Exercise 1.6 Simplifying a Formula (★)

Consider the propositional formula

$$F = ((\neg A \vee \neg B) \wedge \neg A) \wedge ((\neg B \wedge \neg A) \vee C).$$

Give a formula G that is equivalent to F , but in which each atomic formula A , B , and C appears at most once. **Prove** that $F \equiv G$ by providing a sequence of equivalence transformations with **at most** 6 steps.

Expectation. Your proof should be in the form of a sequence of steps, where each step consists of applying the definition of $\rightarrow$ (that is $F \rightarrow G \equiv \neg F \vee G$), one of the rules given in Lemma 2.1 of the lecture notes, or one of the following rules: $F \wedge \neg F \equiv \perp$, $F \wedge \perp \equiv \perp$, $F \vee \perp \equiv F$, $F \vee \neg F \equiv \top$, $F \wedge \top \equiv F$, and $F \vee \top \equiv \top$. Lemma 2.1 states rules involving propositional symbols, but you may apply those rules at the level of formulas (see Section 2.3.5 of the lecture notes). For this exercise, associativity is to be applied as in Lemma 2.1 3). Each step of your proof should apply a **single** rule **once** and state **which** rule was applied.

Questions?